

Optimal Athletic Programming in Triathlon using Linear and Integer Programming

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Three decisions, one athlete

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Model 1 (LP) $\rightarrow$ **Model 2 (0–1 MIP)** $\rightarrow$ **Model 3 (LP)**
training load race calendar pacing strategy

Coupled: Model 1's fitness state gates race readiness and sets the race-day energy budget.

Real data: 15 months of my own training

- Strava export: **598 activities**, 574 kept (Oct 2023 – Jun 2026)
- Per activity: Banister training impulse
$$\text{TRIMP} = t \cdot \text{HRr} \cdot 0.64 e^{1.92 \text{ HRr}}$$
- Exponential smoothing $\Rightarrow$ fitness CTL ($\tau=42\text{d}$),
fatigue ATL ($\tau=7\text{d}$)
- Current state: CTL = 78.1, ATL = 91.5,
TSB = -13.4 (fit but fatigued)
- Intensity buckets vs. *sport-specific* threshold HR
(run 173 / bike 166 / swim 164 bpm)

r_{db}/h	easy	mod	hard
swim	43.5	94.1*	136.6*
bike	68.3	86.9	142.7*
run	81.6	102.8	166.0*

* theoretical (sparse data)

Model 1 — training-load allocation (LP)

Variables: hours $h_{dbw} \geq 0$ per discipline $\times$ bucket $\times$ week ($3 \times 3 \times 16 = 144$)

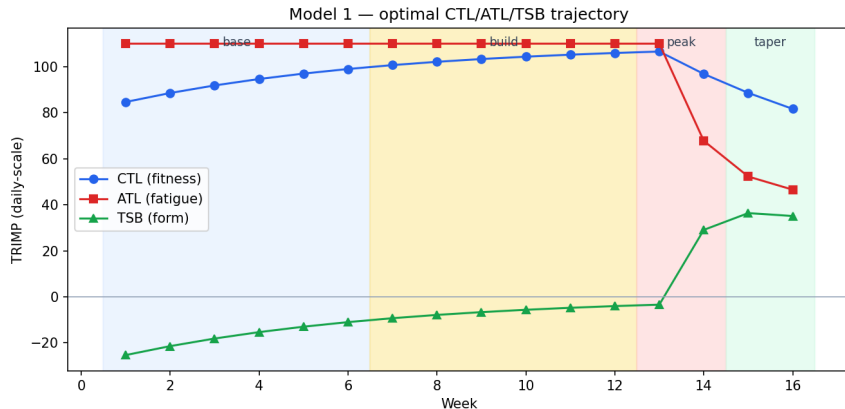
Banister dynamics as linear equalities (exact — smoothing is linear in load):

$$L_w = \sum_{d,b} r_{db} h_{dbw}, \quad \text{CTL}_w = \lambda_c \text{CTL}_{w-1} + (1-\lambda_c) \frac{L_w}{7}, \quad \text{ATL}_w = \lambda_a \text{ATL}_{w-1} + (1-\lambda_a) \frac{L_w}{7}$$

Objective: race-day performance $\max p_T = k_1 \text{CTL}_T - k_2 \text{ATL}_T$ (TSB-max alone would be degenerate: optimum = rest!)

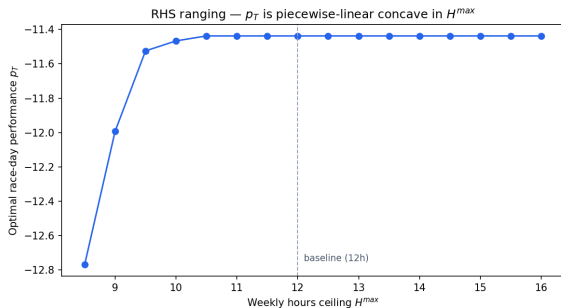
Constraints: weekly hour ceilings (phase-dependent) $\cdot$ discipline min/max $\cdot$ 80/20 polarization $\cdot$
 $\text{ATL} \leq 110 \cdot \text{ramp} \leq +10\%/ \text{week} \cdot \text{taper caps}$

Model 1 — the optimizer discovers periodization



Rides the fatigue ceiling for 13 weeks (CTL 78 $\rightarrow$ 107), then tapers. Only weeks 15–16 are *constrained* to taper — **week 14 tapers by choice**. Race day: TSB = +35.

Model 1 — duality in action



RHS ranging (hours ceiling 8–16h):

- value function piecewise-linear, concave; slopes $1.55 \rightarrow 0.94 \rightarrow 0.12 \rightarrow 0$
 - saturates at **10.5 h/week**: the ATL ceiling binds first
- ⇒ *recovery capacity, not free time, limits this athlete*

Largest shadow prices: race-week minimums ($-12.9/\text{forced run hour in week 16}$) — the model wants full rest

Model 1 — alternative optima & heuristic benchmark

Alternative optima (degeneracy made visible): fix $p_T = p^*$, re-solve min/max total hours

same performance with **124.2h** or **162.3h** of season training — a 38h efficiency gap

vs. standard coaching heuristic (fill available hours, 20/45/35 split, 80/15/5 intensity):

- violates the fatigue ceiling in **13 of 16 weeks** (chronic overtraining)
- $p_T = -69.5$ vs. optimal -11.4
- but: heuristic race-day TSB $+21$ is inside the coaching band $+15..25$; the LP's $+35$ is not $\rightarrow$ see k_2 sensitivity

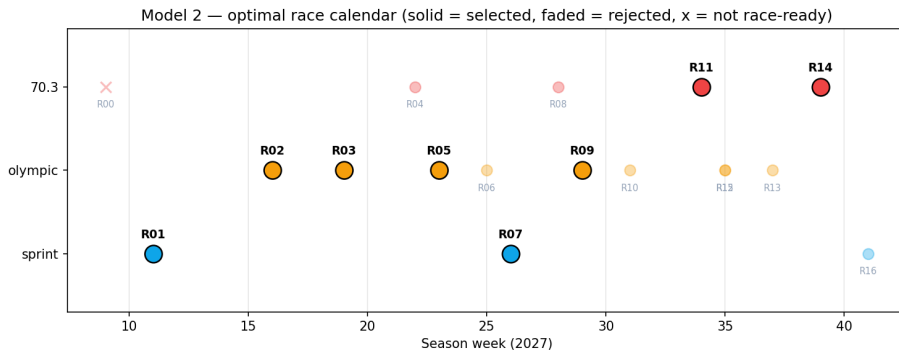
Model 2 — race-calendar selection (0–1 MIP)

17 candidate races (Greek / European / long-haul), binary x_i

$$\max \sum_i q_i \underbrace{(\alpha \text{ pts}_i + \beta \text{ strat}_i + \gamma \text{ enj}_i + \varphi \text{ fit}_i)}_{\text{weather-risk-adjusted value}} x_i$$

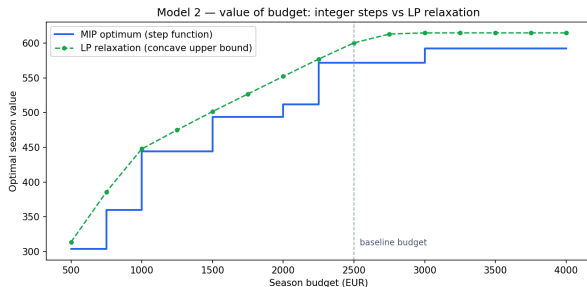
- **Two-dimensional knapsack:** $\sum \kappa_i x_i \leq 2500$ EUR and $\sum \delta_i x_i \leq 12$ vacation days
- recovery spacing $x_i + x_j \leq 1$ (1/2/4 weeks by distance class); ≤ 2 races/month; ≥ 1 A-race
- **readiness gate from Model 1:** projected CTL must reach the class requirement — excludes a high-value early-season 70.3
- suitability fit_i from *objective* course data: elevation gain/km vs. athlete's flat-course preference

Model 2 — optimal season



8 of 17 races, value 571.7, **2250/2500 EUR** and **11/12 days used**. Recovery forces National Champs vs. home 70.3 — optimizer picks the 70.3.

Model 2 — why Branch & Bound is needed



- LP relaxation: $z = 600.5$ vs. MIP 571.7 — **integrality gap 5.0%**
- 8 fractional variables; the champs/70.3 conflict splits exactly **0.5/0.5**
- MIP value-of-budget is a **step function** under the concave LP curve
- above 3000 EUR the *vacation-days* budget binds instead — the scarce resource switches
- LP duals: 0.094 value/EUR, ≈ 0 per day off

Model 2 — network-flow extension

Drop the knapsacks and monthly cap $\Rightarrow$ only recovery spacing remains:

feasible calendars = paths in a DAG (nodes: races in week order; arc $i \rightarrow j$ iff $\text{gap} \geq \text{recovery}(i)$)

Three methods, identical result ($z = 639.4$):

- 1 dynamic programming (longest path, topological order)
- 2 **flow LP with continuous variables** — solution integral anyway: network matrix $\Rightarrow$ *totally unimodular*
- 3 restricted 0–1 MIP

Node–arc incidence matrix $\Rightarrow$ every basis is a spanning tree: exactly the class *Network Simplex* is designed for (18 nodes, 136 arcs). $\Rightarrow$ the knapsack constraints are precisely what *breaks* the network structure and forces Branch & Bound in the full model.

Model 3 — race pacing (LP with linearization)

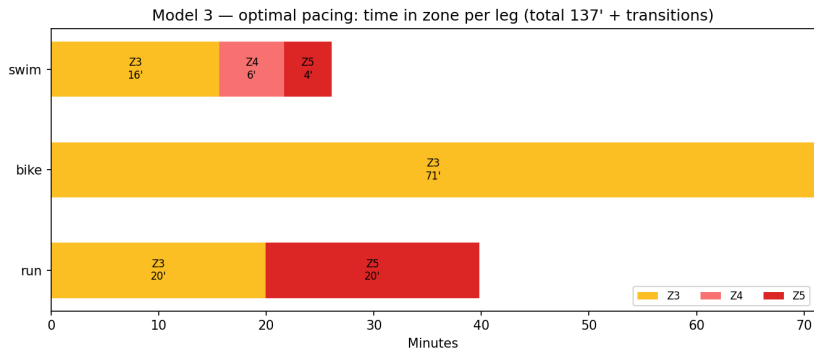
Variables: minutes $t_{\ell z}$ per leg $\times$ intensity zone (Z1–Z5); 15 variables

Linearization: within a zone, speed and energy rate are constants — bike speed $\propto (\text{power})^{1/3}$ (aero drag), economy degrades with intensity

$$\min \sum_{\ell, z} t_{\ell z} + T_1 + T_2 \quad \text{s.t.} \quad \underbrace{\sum_z s_{\ell z} t_{\ell z} \geq d_\ell}_{\text{distances}}, \underbrace{\sum e_{\ell z} t_{\ell z} \leq k_E \cdot \text{CTL}_T}_{\text{energy budget (Model 1)}}, \underbrace{\text{Z4+ time caps}}_{\text{per leg}}, \underbrace{\varphi \cdot G}_{\text{bike} \rightarrow \text{run}}$$

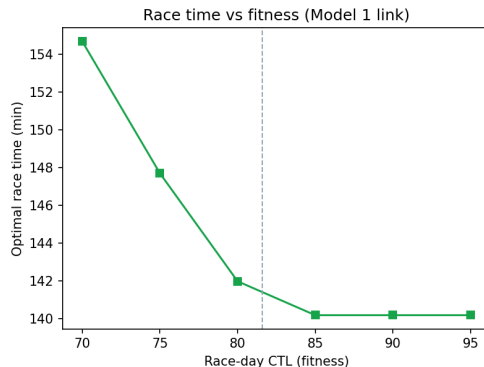
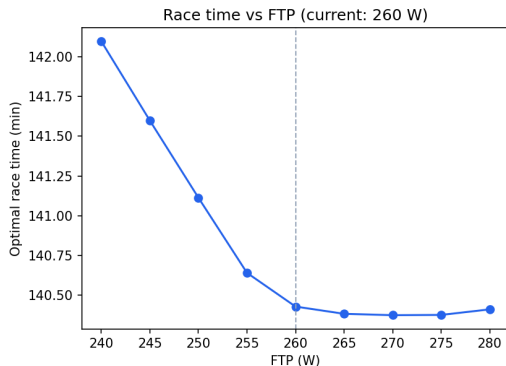
The coupling: each kJ of hard bike work reduces effective run coverage.

Model 3 — “steady bike, hard run,” derived



Olympic distance in 140.4 min; 9389 kJ budget **exactly** exhausted. Bike entirely at Z3: hard biking buys little speed (cube root) and drains the run.

Model 3 — what is a watt worth?



Energy dual: $1 \text{ kJ} = 0.43 \text{ s}$. Below 260 W each watt $\approx 6 \text{ s}$; **at the current FTP the value collapses** — with a fixed energy budget, more power cannot be fueled. One CTL point = 69–84 s up to CTL ≈ 85 . $\Rightarrow$ *build endurance, not peak power.*

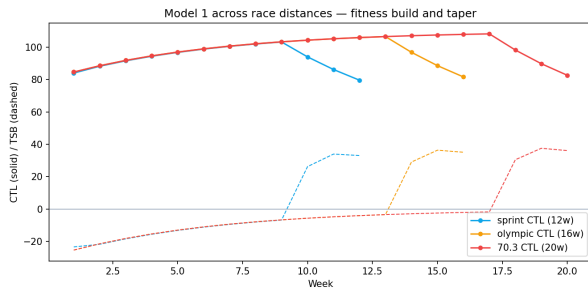
Model 3 — validated against three real races

Race	CTL	Swim		Bike		Run	
		mdl	act	mdl	act	mdl	act
Spetsathlon '26 (sprint*)	59.0	12.9	13.2	44.4	50.9	18.9	17.7
Epidavros '25 (Olympic)	55.6	27.6	25.8	83.7	91.5	43.1	42.0
70.3 Costa Navarino '25 (fueled)	71.4	33.3	32.9	181.1	180.3	86.3	115.2

Race-day fitness reconstructed from the pipeline; $k_E = 142$ calibrated on Epidavros. *non-standard course.

- swim always within 0.4 min; 70.3 bike within **one minute**
- **binding constraints switch with distance**: speed caps (sprint) $\rightarrow$ energy (Olympic) $\rightarrow$ *infeasible unfed* (70.3)
- fueling stays LP — fixed rate or *per-leg decision* $F_\ell \leq \bar{r}_\ell \cdot \text{leg time}$; calibrated $r \approx 21$ kJ/min = **75 g carbs/h, the standard guideline**; absorption duals price **gut training** (+1 kJ/min bike ≈ 1.7 min at 70.3)
- the 70.3 run “wall” (4:53 $\rightarrow$ 5:52/km, HR 162 $\rightarrow$ 145): the energy constraint violated in real life — mechanism right, within-leg timing beyond the model

Sensitivity across problem sizes and parameters



Distance scenarios (12/16/20 weeks, 144–240 vars): longer horizons help (p_T : $-13.5 \rightarrow -10.4$); build–taper structure invariant

k_2 **fatigue weight** $\in [1, 3]$: race-day TSB stays 35 ± 2 — taper depth is *constraint*-determined (race-week minimums), robust to the objective's calibration

Cost ranging (c -perturbation): plan invariant for $k_1 \in [0.68, 1.56]$, $k_2 \in [1.29, 2.95]$ — robust to the least measurable parameters

Sizes/times: LP 192 vars / MIP 17 bins / LP 15 vars; all <10 ms (CBC/Simplex)

Model 2 — the dual problem

LP relaxation is already in standard form $\max\{c^T x \mid Ax \leq b, x \geq 0\}$, so the dual is $\min\{b^T y \mid A^T y \geq c, y \geq 0\}$:

$$\kappa_i y_B + \delta_i y_D + y_{\text{month}(i)} + \sum_j y_{ij} - [i \in A] y_A \geq v_i$$

a race enters only if its value covers the shadow cost of what it consumes

Dual	Meaning	Price
$y_{13,14}$	Hellenic Champs vs. Costa Navarino conflict	56.40
$y_{06,07}$	Hamburg vs. Thessaloniki conflict	31.67
y_B	one extra euro	0.094
y_D	one extra vacation day	≈ 0

⇒ the scarce resource is neither money nor days off but **calendar space**: recovery between races.

Limitations — stated honestly

- **Weekly time step** for Banister dynamics (daily would shift constants, not structure)
- **Sequential pipeline**, not joint optimization — co-optimizing training + calendar is a natural (larger) MIP extension
- **Constructed parameters** in Models 2–3 (race values, zone speeds, k_E) — documented assumptions; sensitivity analysis shows which conclusions survive perturbation
- LP's +35 TSB over-taper traced to race-week minimum constraints, not to the model concept

- Three coupled models on **real personal data**, spanning the syllabus: LP + Simplex, duality & sensitivity, IP + Branch & Bound, network flows + total unimodularity
- The optimizer **re-derives coaching wisdom**: periodization, polarized intensity, “steady bike, hard run”
- Duals price the athlete’s true bottlenecks: *recovery capacity* (training), *money* (calendar), *endurance* — *not watts* (race day)
- Fully reproducible: `python -m src.run_all`, 44 tests
<https://github.com/manos1503/triathlon-optimization>

Thank you! Questions?